STAT301 Worksheet 7 Name

1. Let A and B be the following matrices, calculate A*B, and B*A. Are they the same?

$A = \begin{matrix} 5 & 1 \\ 3 & -2 \end{matrix}$ $B = \begin{matrix} 2 & 0 \\ 4 & 3 \end{matrix}$

$5 \cdot 2 + 1 \cdot 4$ $5 \cdot 0 + 1 \cdot 3$ 14 5

$3 \cdot 2 + -2 \cdot 4$ $3 \cdot 0 + -2 \cdot 3$ -2 -6

2. Let A and B be the following matrices, is it possible to find A*B?

$A = \begin{matrix} 3 & 1 & 2 \\ 0 & 2 & -1 \\ 1 & 0 & 0 \end{matrix}$ $B = \begin{matrix} 1 & 4 & 3 \\ -2 & 0 & 3 \end{matrix}$

don't work

No A is 3x3 and B is 2x3

3. Let A and B be the following, calculate $(A * B)^T$ and $B^T * A^T$ and $A^T * B^T$, which two of them are equal?

$A = \begin{matrix} 1 & 1 & 1 \\ 2 & 3 & 5 \end{matrix}$ $B = \begin{matrix} -5 & 2 \\ 1 & 0 \\ 0 & 1 \end{matrix}$

$(A \cdot B)^T$

$-5 + 1 + 0$ $2 + 0 + 1$

$-10 + 3 + 0$ $4 + 0 + 5$

$\begin{bmatrix} -4 & 3 \\ -7 & 9 \end{bmatrix}^T = \begin{bmatrix} -4 & -7 \\ 3 & 9 \end{bmatrix}$

$\begin{matrix} -5 & 10 \\ 2 & 0 & 1 \end{matrix} \cdot \begin{matrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{matrix}$

$-5 + 1 + 0$ $-10 + 3 + 0$

$2 + 0 + 1$ $4 + 5 + 0$

$\begin{bmatrix} -4 & -7 \\ 3 & 9 \end{bmatrix}$

$A^T \cdot B^T$

$\begin{matrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{matrix}$ $\begin{matrix} -5 & 10 \\ 2 & 0 & 1 \end{matrix}$

$-5 + 4$

$-5 + 6$

$-5 + 10$

$\begin{bmatrix} -11 & 2 \\ 1 & 3 \\ 5 & 5 \end{bmatrix}$

$(A \cdot B)^T = B^T \cdot A^T$

4. Find the determinant of B if

$$B = \begin{matrix} 3 & 4 & 0 \\ 0 & 2 & 3 \\ 1 & 1 & 0 \end{matrix}$$

$(-1)^{3+1}(1)\det\begin{pmatrix} 4 & 0 \\ 2 & 3 \end{pmatrix} + (-1)^{3+2}(1)\det\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} + 0$

$1 \cdot 4 \cdot 3 - 0$ $\quad -1 \cdot 9$

$12 + -9 = \boxed{3}$

5. Find A^{-1} if

$1 \cdot 1 - 2 \cdot 1 = -1$

$$A = \begin{matrix} 1 & 1 \\ 2 & 1 \end{matrix}$$

$\frac{1}{-1} = -1\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$

6. If $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$, find the eigenvalues and unit eigenvectors.

$\det(A - \lambda I) = 0$

$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$ $\quad \det\begin{bmatrix} 0-\lambda & 1 \\ -2 & -3-\lambda \end{bmatrix}$

$(0-\lambda)(-3-\lambda) - 2$

$\lambda^2 + 3\lambda + 2 = 0$

$(\lambda - 1)(\lambda - 2)$

$(A - \lambda I)_x = 0$

$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$1x_1 \quad 1x_2 = 0$

$x_1 + x_2 = 0$

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

$2x_1 + 1x_2 = 0$

$-2x_1 + -1x_2 = 0$

$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$